QUANT

Averages, LCM & Decimals

eBOOK

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Fractions

Fractions play an important role in most of the competitive exams.

Following are some definitions and fundamentals one needs to know before starting the topic of Fractions.

- Numbers of the form 3/4, 4/5, are called fractions. A fraction can be written as p/q where $q \neq 0$.
- If the numerator and denominator of a fraction are multiplied / divided by the same number then the value of the fraction does not change.
- For any positive proper fraction p/q (p<q), the value of the fraction increases when both the denominator and numerator are added by the same positive number. e.g $\frac{3}{4}$ = 0.75, $\frac{3+1}{4+1} = \frac{4}{5}$ = 0.8.
- For any positive proper fraction p/q (p<q), the value of the fraction decreases when both the numerator and denominator are subtracted by the same positive number. e.g $\frac{3}{4}$ = 0.75, $\frac{3-1}{4-1} = \frac{2}{3}$ = 0.67
- For any positive improper fraction p/q (p>q), the value of the fraction decreases when both the numerator and the denominator are added to the same positive. e.g. $\frac{5}{4}$ = 1.25, adding 1 to the numerator and the denominator, we get $\frac{5+1}{4+1} = \frac{6}{5}$ = 1.2, which is less than 1.25.
- For any positive improper fraction p/q (p>q), the value of the fraction increases when both the numerator and denominator are subtracted by the same positive number. e.g. $\frac{5}{4}$ = 1.25, by subtracting 1 from both the numerator and denominator we get, $\frac{5-1}{4-1} = \frac{4}{3}$ = 1.33 > 1.25.

1.1 Types of Fractions

Common Fractions: Fractions such 3/4, 32/43 etc are called common or vulgar fractions.
Decimal Fractions: Fractions whose denominators are 10, 100, 1000, ... are called decimal fractions.
Proper Fraction: A fraction whose numerator is less than its denominator is known as a proper fraction e.g. 3/4
Improper Fraction: A fraction whose numerator is greater than its denominator is known as an improper fraction. e.g. 4/3
Mixed Fractions: Fractions which consists of an integral part and a fractional part are called mixed fractions. All improper fractions can be expressed as mixed fractions and vice versa. e.g. $1\frac{3}{4}$.
Recurring Decimals: A decimal in which a set of figures is repeated continually is called a recurring or periodic or a circulating decimal. e.g. $1/7$ = 0.142857…….. the dots indicate that the figure between 1 and 7 will repeat continuously.

1.2 Addition of Fractions

Mixed fractions can be added by adding the integral and fractional part separately or by converting them into improper fractions. e.g. $3\frac{4}{5} + \frac{11}{12}$ can be added as either $3 + \frac{4}{5} + \frac{11}{12} \Rightarrow 3 + \frac{4\times12+11\times5}{60}$ $\Rightarrow 3 + \frac{103}{60} \Rightarrow 3 + 1\frac{43}{60} = 4\frac{46}{60} = \frac{283}{60}$ or by converting the mixed fractions into improper fraction, $\frac{19}{5} + \frac{11}{12} = \frac{(19\times12+11\times5)}{60} = \frac{283}{60}$.

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1.3 Decimals and Fractions to be remembered

You get many questions in the exams based on Percentage, Profit, Interest etc. in which you have to calculate, say 87.5 % of 800, 58.33 % of 2400 etc. Calculating these values with the help of traditional methods is time-consuming. IF you have the fraction approach, you can crack these easily i.e., if you know that 87.5 % is just 7/8th of the number and 58.33 % is 7/12th of the number, then it becomes easy to calculate.

% age	Fraction		% age	Fraction
50 %	$1/2$		$55\,5/9$ %	$5/9$
$33\,1/3$ %	$1/3$		$77\,7/9$ %	$7/9$
$66\,2/3$ %	$2/3$		$88\,8/9$ %	$8/9$
25 %	$1/4$			
75 %	$3/4$		$9\,1/11$ %	$1/11$
20 %	$1/5$		$18\,2/11$ %	$2/11$
40 %	$2/5$		$27\,3/11$ %	$3/11$
60 %	$3/5$		$36\,4/11$ %	$4/11$
80 %	$4/5$		$45\,5/11$ %	$5/11$
			$54\,6/11$ %	$6/11$
$16\,2/3$ %	$1/6$		$63\,7/11$ %	$7/11$
$83\,1/3$ %	$5/6$		$72\,8/11$ %	$8/11$
$14\,2/7$ %	$1/7$		$81\,9/11$ %	$9/11$
			$90\,10/11$ %	$10/11$
$12\,1/2$ %	$1/8$			
$37\,1/2$ %	$3/8$		$8\,1/3$ %	$1/12$
$62\,1/2$ %	$5/8$		$41\,2/3$ %	$5/12$
$87\,1/2$ %	$7/8$		$58\,1/3$ %	$7/12$
			$91\,2/3$ %	$11/12$
$11\,1/9$ %	$1/9$		$6\,2/3$ %	$1/15$
$22\,2/9$ %	$2/9$		$6\,1/4$ %	$1/16$
$44\,4/9$ %	$4/9$		5 %	$1/20$

Anything doubles to increase by 100 % and becomes 200 %.
Anything trebles to increase by 200 % and becomes 300 %.

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1.4 Solved Examples based on Fractions and Decimals:

Ex. 1: One-quarter of one-seventh of a land is sold for Rs.30, 000. What is the value of an eight thirty-fifths of land?

Sol: One-Quarter of one-seventh $=\frac{1}{4}\times\frac{1}{7}=\frac{1}{28}$. Now, $\frac{1}{28}$ of a land will cost = Rs.30,000

$\therefore \frac{8}{35}$ of the land will cost $\frac{30{,}000\times 28\times 8}{35}$ = Rs.1,92,000

Ex. 2: After taking out of a purse $\frac{1}{5}$ of its contents, $\frac{1}{12}$ of the remainder was found to be Rs7.40. What sum did the purse contain at first?

Sol: After taking out $\frac{1}{5}$ of its contents, the purse remains with $\frac{4}{5}$ of contents.

Now $\frac{1}{12}$ of $\frac{4}{5}$ = Rs. 7.40 or $\frac{1}{15}$ = Rs. 7.40 $\therefore$ 1 = Rs.111.

Ex. 3: A sum of money increased by its seventh part amounts to Rs. 40. Find the sum.

Sol: As per the question, S+ $\frac{S}{7}$ = Rs.40

$\Rightarrow \frac{8S}{7}$ = Rs. 40 $\Rightarrow$ S = Rs.35.

Ex. 4: A train starts full of passengers. At the first station, it drops one-third of these and takes in 96 more. At the next, it drops one – half of the new total and takes in 12 more. On reaching the next station, there are found to be 248 left. With how many passengers did the train start?

Sol: Let the train start with x passengers. After dropping one –third and taking in 96 passengers, the train has x - $\frac{x}{3}+96=\frac{2x}{3}+96=\frac{2x+288}{3}$ passengers.

At the second station, the number of passengers $=\frac{2x+288}{6}$+12. Now, $\frac{2x+288}{6}$+12=248

Or,2x + 288 = 1416 $\therefore$ x = 564.

Ex. 5: A motorcycle, before overhauling, requires 5/6 hour service time every 90 days, while after overhauling. It requires 5/6 hour service time every 120 days. What fraction of the pre-overhauling service time is saved in the latter case?

Sol: LCM of 90 and 120 = 360. So, in 360 days, the pre-overhauling service time$=\frac{5}{6}\times\frac{360}{90}=\frac{10}{3}$ hrs.

And after overhauling, the service time $=\frac{5}{6}\times\frac{360}{120}=\frac{5}{2}hrs$. Time Saved = $\frac{10}{3}-\frac{5}{2}=\frac{5}{6}hrs$ hrs

$\therefore$ The required answer = $\frac{\frac{5}{6}}{\frac{10}{3}}=\frac{5}{6}\times\frac{3}{10}=\frac{1}{4}$

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Square Root

To find the square root of a number, there are two methods available. First method is prime factorization and the second is the conventional square root method. Factorization is suitable only, when the numbers are relatively small and their factors can be easily found. Considering the kind of questions which appear in the competitive exams, firstly we are going to learn the conventional square root calculation method. In this, firstly the number is divided into pairs from the right hand side. If in the beginning there is a pair, then the starting is done with that pair, and if there is a single digit-number, then that would be the starting point.

2.1 Method to find Square Root:

Illustration 1: Find the square root of 64516.

Sol:

```
        2  5   4
2     | 6 45 16
      | 4
45    | 2 45
      | 2 25

504   | 20 16
      | 20 16
      | X
```

In the first place, when 6 is the single number left, after making pairs from the RHS, then you should take a number, which can be multiplied by the same number itself, and the result is less than equal to 6, which is 2.
After subtracting 4 from it, the new pair 45 is taken. The number now becomes 245. The previous quotient is doubled and 4 is obtained. Then a number 'x' is written with 4, in such a way that the product of '4x' and 'x' is less than or equal to 245. So value of x is 5.
The remainder in the next step is 20 and the last pair 16 is written with it.
Then the previous quotient 25 is doubles and 50 is obtained and a number 'y' is written with it in such a way that the product of '50y' and 'y' is less than or equal to 2016.
When y is substituted by 4 the product is 2016.
In this way, the final quotient 254 is the square root of 64,516.

Illustration 2: Find the square root of 328329.

Sol:

```
        5  7  3
5     | 32 83 29
      |   25
107   |  7 83
      |  7 49

1143  |  34 29
      |  34 29
      |    x
```

Applying the square root calculation method, in the first place after making pairs from the RHS, you are left with the number 32. Now you should take a number, which can be multiplied by the same number itself, and the result is less than equal to 32, which is 5.
After subtracting 25 from it, the new pair 83 is taken. The number now becomes 783. The previous quotient is doubled and 10 is obtained. Then a number 'x' is written with 10, in such a way that the product of '10x' and 'x' is less than or equal to 783. So value of x is 7.

The remainder in the next step is 34 and the last pair 29 is written with it.

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Then the previous quotient 57 is doubled and 114 is obtained and a number 'y' is written with it in such a way that the product of '114y' and 'y' is less than or equal to 3429.
When y is substituted by 3 the product is 3429.
In this way, the final quotient 573 is the square root of 328,329.

2.2 Solved Examples based on Square Root:

Ex. 1: Find the smallest number with which 60 should be multiplied, so that it becomes a perfect square.

Sol: In order to answer such questions, firstly the prime factorization of the number is done. The factors of 60 are 2 × 2 × 3 × 5.
In this it can be seen that '2' is occurring twice, but 3 and 5 are occurring only once. In order to make a number a perfect square every prime factor should be there twice or an even number of times. So a '5' and '3' is required, the product of which is 15. Therefore 15 is the smallest number.

Ex. 2: In a class, each of the students contributed as many paise as there are number of students. If the total collection was Rs. 144, what is the number of students in the class?

Sol: Let the number of students in the class be x. Now each of these students contributed 'x' paise each. So the total collection will be x^2 paise.
Now the total collection is given to be Rs. 144, which is 14400 paise.
As per the statement of the question $x^2 = 14400$
$\Rightarrow$ x = 120. Thus there are 120 students.

Ex. 3: What is the value of $\sqrt{10+\sqrt{25+\sqrt{108+\sqrt{154+\sqrt{225}}}}}$?

Sol: Given exp.

$$= \sqrt{10+\sqrt{25+\sqrt{108\sqrt{154+15}}}}$$

$$= \sqrt{10+\sqrt{25+\sqrt{108+\sqrt{169}}}}$$

$$= \sqrt{10+\sqrt{25+\sqrt{108+13}}} = \sqrt{10+\sqrt{25+\sqrt{121}}}$$

$$= \sqrt{10+\sqrt{25+11}} = \sqrt{10+\sqrt{36}} = \sqrt{10+6} = \sqrt{16} = 4$$

Ex. 4: What is the value of $\sqrt{175+\sqrt{2401}}$?

Sol: Given exp. = $\sqrt{176+49} = \sqrt{225} = 15$

Ex. 5: What is the value of $\left(\frac{\sqrt{625}}{11} \times \frac{14}{\sqrt{25}} \times \frac{11}{\sqrt{196}}\right)$?

Sol: Given exp. = $\frac{25}{11} \times \frac{14}{5} \times \frac{11}{14} = 5$

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Ex. 6: What is the value of $\left(\sqrt{\frac{225}{729}}-\sqrt{\frac{25}{144}}\right)\div\sqrt{\frac{16}{81}}$?

Sol: Given exp. =

$$\left(\frac{\sqrt{225}}{\sqrt{729}}-\frac{\sqrt{25}}{\sqrt{144}}\right)+\frac{\sqrt{16}}{\sqrt{81}}=\left(\frac{15}{27}-\frac{5}{12}\right)+\frac{4}{9}=\left(\frac{15}{108}\times\frac{9}{4}\right)=\frac{5}{16}$$

Ex. 7: What is the digit in the unit's place in the square root of 15876?

Sol: $\sqrt{15876}=126$. Hence unit digit in the square root of 15876 is 6.

Ex. 8: What is the value of $\sqrt{0.01}+\sqrt{0.81}+\sqrt{1.21}+\sqrt{0.0009}$?

Sol: Given exp. = $\sqrt{\frac{1}{100}}+\sqrt{\frac{81}{100}}+\sqrt{\frac{121}{100}}+\sqrt{\frac{9}{10000}}$

$$=\frac{1}{10}+\frac{9}{10}+\frac{11}{10}+\frac{3}{100}$$

= 0.1 + 0.9 + 1.1 + 0.03 = 2.13

Ex. 9: What is the value of $\sqrt{0.0025}\times\sqrt{2.25}\times\sqrt{0.0001}$?

Sol: Given exp. = $\sqrt{\frac{25}{10000}}\times\sqrt{\frac{225}{100}}\times\sqrt{\frac{1}{10000}}=\frac{5}{100}\times\frac{15}{10}\times\frac{1}{100}$

$$=\frac{75}{100000}=0.00075$$

Ex. 10: What is the value of $\sqrt{1.5625}$?

Sol: $\sqrt{1.5625}=1.25$

Ex. 11: If $\sqrt{0.00000676}=0.0026$, then what is the square root of 67,60,000?

Sol: $\sqrt{6760000}=\sqrt{0.00000676\times10^{12}}$

$$=\sqrt{0.00000676}\times\sqrt{10^{12}}=0.0026\times10^{6}=2600$$

Ex.12: If $\sqrt{18225}=135$, then what is the value of $\left(\sqrt{182.25}+\sqrt{1.8225}+\sqrt{0.018225}+\sqrt{0.00018225}\right)$?

Sol: Given exp.

$$=\sqrt{\frac{18225}{10^2}}+\sqrt{\frac{18225}{10^4}}+\sqrt{\frac{18225}{10^6}}+\sqrt{\frac{18225}{10^8}}$$

$$=\frac{\sqrt{18225}}{10}+\frac{\sqrt{18225}}{10^2}+\frac{\sqrt{18225}}{10^3}+\frac{\sqrt{18225}}{10^4}=\frac{135}{10}+\frac{135}{100}+\frac{135}{1000}+\frac{135}{10000}$$

= 13.5 + 1.35 + 0.135 + 0.0135 = 14.9985

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Ex. 13: An Armyman wants to arrange his men in the form of a perfect square, but he finds there are 52 men too many. What will be the total number of men in front row, if the total number of men with him is 14693?

Sol: Required number of men in the front row = $\sqrt{14693-52} = \sqrt{14641} = 121$.

Note: No perfect square ends with 2, 3, 7, 8, and odd number of zeroes i.e. any number, which has 2, 3, 7 and 8 at its unit's place and any number ending with odd number of zeroes can never be a perfect square.

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Cube & Cube Root

Before learning how to find the cube root, let us first learn the meaning of cube. The process of cubing a number is multiplying the number three times. The exponent used for cubes is 3, which is also denoted by the superscript^3 e.g. the cube of 4 will be calculated as $4 \times 4 \times 4 = 4^3 = 64$ or $8^3 = 8 \times 8 \times 8 = 512$.

Now in order to find the cube root of a number the only method available is prime factorization. The approach applied is that each of the number, which is a perfect cube, will have every prime factor appearing in a group of 3. This is done because unlike square root there is no other conventional method to find the cube root. After the prime factorization, each of the prime factors is selected once for every three times it is appearing in the number. You will understand the concept better with the help of the following examples.

3.1 Solved Examples based on cubes and cube root:

Ex.1: Find out the cube root of 1728.

Sol: First we will do prime factorization.
Prime factorization of 1728 is = $2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 3 \times 3 \times 3$
$= (2 \times 2 \times 3) \times (2 \times 2 \times 3) \times (2 \times 2 \times 3)$. $= 12 \times 12 \times 12 \Rightarrow$ the cube root of 1728 is 12.

Ex. 2: Find the cube root of 9261.

Sol: The factors of 9261 are $3 \times 3 \times 3 \times 7 \times 7 \times 7$. $= (7 \times 3) \times (7 \times 3) \times (7 \times 3)$
$= 21 \times 21 \times 21 \Rightarrow$ the cube root of 9261 is 21.

Ex. 3: Find the cube root of 15625.

Sol: The factors of 15625 are $= 5 \times 5 \times 5 \times 5 \times 5 \times 5$. $= (5 \times 5) \times (5 \times 5) \times (5 \times 5)$
$=25 \times 25 \times 25 \Rightarrow$ the cube root of 15625 is 25.

Ex. 4: Find the smallest number with which 43904 should be multiplied so as to make it a perfect cube.

Sol: The prime factorization of 43904 is $2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 7 \times 7 \times 7$. As you can see 2 appears in this 7 times, so groups of three 2s can be made 2 times and still one 2 will be left and 7 appears in a group of three only. Thus the groups can be made as $\underline{2 \times 2 \times 2} \times \underline{2 \times 2 \times 2} \times 2 \times \underline{7 \times 7 \times 7}$ and as you see a single 2 is left out and now it needs two more 2s to make it a perfect cube i.e. it should be multiplied by 4.

Ex. 5: Find the smallest number by which 73002 should be divided so as to make it a perfect cube.

Sol: The prime factorization of the number 73002 is $\underline{23 \times 23 \times 23} \times 2 \times 3$. As you can see 23 already appears three times, there is a single 2 and a single 3. This implies if this number is divided by 6, it will become a perfect cube. Hence the answer to the question will be 6.

Ex. 6: Find the cube and cube root of 27.

Sol: The cube of 27 will be found by multiplying it three times i.e. $27 \times 27 \times 27 = 19683$.
Now the prime factorization of 27 is $3 \times 3 \times 3$. As you know, you need to take one number for every group of three. As 3 appears three times, taking a single 3 for the same, the cube root of 27 is 3.

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LCM & HCF

Least Common Multiple - LCM

The least common multiple (LCM) of two or more numbers is the smallest of the numbers, which is exactly divisible by each of them.
e.g Consider two number 12 and 15
Multiples of 12 are: 12, 24, 36, 48, 60, 72, 84, 96, 108, 120, 132, ...
Multiples of 15 are: 15, 30, 45, 60, 75, 90, 105, 120, 135, ...
The common multiples of both 12 and 15 are 60, 120, 180, ...
The least common multiple is 60.

How to find the LCM of two or more numbers?
The LCM of two numbers can be found by the product of the factors of the two numbers after eliminating repetition of the common factors.
In the above Ex., the common factor for 12 and 15 are 3.
Therefore, the LCM will be 3*4*5 = 60.
Alternatively, LCM is the product of all prime factors of the given numbers, the common factors among them being in their highest degree. e.g, The LCM of 5x2y3z5 and 3xy2z7 will be 5*3*x2y3z7 = 15x2y3z7, where x, y and z are the prime factors.

Greatest Common Divisor (GCD)/Highest Common Factor (HCF):

The highest common factor of two or more numbers is the greatest number, which divides each of those numbers exact number of times. e.g. HCF of 24 and 36 is 12.

How to find the HCF of two or more numbers?

a) Express the two numbers as product of prime numbers separately.
b) Take the product of prime numbers common to both numbers.

4.1 LCM and HCF of Fractions

LCM of fractions = $\dfrac{\text{LCM of numerators}}{\text{HCF of denominators}}$;

e.g. LCM of $\dfrac{3}{4}$ and $\dfrac{1}{2}$ = $\dfrac{3\,(\text{LCM of numerators})}{2\,(\text{HCF of denominators})}$

HCF of fractions = $\dfrac{\text{HCF of numerators}}{\text{LCM of denominators}}$

e.g. HCF of $\dfrac{3}{4}$ and $\dfrac{1}{2}$ = $\dfrac{1(\text{HCF of numerators})}{4(\text{LCM of denominators})}$

Note that the product of the two fractions is always equal to the product of LCM and HCF of the two fractions.

The product of the two fractions = $\dfrac{3}{4} \times \dfrac{1}{2} = \dfrac{3}{8}$.

The product of the LCM and HCF = $\dfrac{3}{2} \times \dfrac{1}{4} = \dfrac{3}{8}$.

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4.2 Important Points in LCM and HCF

- In case of HCF, if some remainders are given, then first those remainders are subtracted from the numbers given and then their HCF is calculated.
- In case of LCM, if a single remainder is given, then firstly the LCM is calculated and then that single reminder is added in that.
- In case of LCM, if for different numbers different remainders are given, then the difference between the number and its respective remainder will be equal. In that case, firstly the LCM is calculated, then that common difference between the number and its respective remainder is subtracted from that.
- Sometimes in case of HCF questions, the same remainder is required is given and the remainder is not given.

4.3 Solved Examples based on LCM and HCF:

Ex. 1: Find the greatest number which when divides 259 and 465 leaves remainders 4 and 6 respectively.

Sol: Here the numbers 259 and 465 leave the remainders 4 and 6 respectively. So the required number will be obtained by finding the H.C.F. of 259 – 4 = 255 and 465 – 6 = 459. The HCF of 255 and 459 is 51 which is the number.

Ex. 2: Find the least number which when divided by 6, 14, 18 and 22 leaves remainder 4 in each case.

Sol: The LCM of 6, 14, 18 and 22 is 1386. In order to get remainder 4 in each case, we will add 4 to the LCM. So the number is 1386 + 4 = 1390.

Ex. 3: Find the least number which when divided by 8, 12, 20 and 36 leaves remainders 6, 10, 18 and 34 respectively.

Sol: Here the numbers are 8, 12, 20 and 36 and the respective remainders are 6, 10, 18 and 34. The difference between numbers and the respective remainders is equal to 2. So first of all find the LCM of 8, 12, 20 and 36 which is 360. The required number is 360 – 2 = 358.

Ex. 4: Fine the greatest number, which when 41, 71 and 91 leaves the same remainder in each case.

Sol: Take the difference between all the three pairs of numbers and their HCF will be the answer i.e. 91 – 41 = 50, 71 – 41 = 30, 91 – 71 = 20. Now the HCF of 50, 30 and 20 is 10.
Sometimes in such questions the common remainder can also be asked.
You can divide any of the numbers given by HCF (91 ÷ 10) and find the remainder to be equal to 1.
Before we switch over to the next topic, it is better to remember the following points.

Ex. 5: Find the L.C.M of $\frac{2}{5}, \frac{3}{10}$ and $\frac{6}{25}$.

Sol: L.C.M. of $\frac{2}{5}, \frac{3}{10}$ and $\frac{6}{25} = \frac{\text{L.C.M. of } 2,3 \text{ and } 6}{\text{H.C.F. of } 5,10 \text{ and } 25}$

L.C.M. of 2, 3 and 6 = 6; H.C.F. of 5, 10 and 25 = 5. Thus the LCM of these three fractions will be 6/5.

Ex. 6: How often will five bells toll together in one hour if they start together and toll at intervals of 5, 6, 8, 12, 20 seconds, respectively?

Sol: The time after which the bells will ring together is the L.C.M. of 5, 6, 8, 12 and 20 seconds, i.e. 120 seconds. The number of times they will toll together in one hour = (3600 ÷ 120) = 30. Thus they will toll together 30 times in an hour.

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⇒Sometimes the question is how many times they toll together in the first hour, in that case after finding the answer like above, you need to add 1 for the start together as well i.e. in the first hour it is 1 more than the usual number of times.

Ex. 7: Find the greatest number of four digits which when divided by 10, 15, 21 and 28 leaves 4, 9, 15 and 22 as remainders, respectively?

Sol: In case of LCM, if for different numbers different remainders are given, then the difference between the number and its respective remainder will be equal. In that case, firstly the LCM is calculated, LCM of 10, 15, 21, and 28 = 420. Now the greatest four-digit number i.e. 9999, when divided by 420 gives remainder 339 therefore, 9999 – 339 = 9660 is the greatest four – digit number divisible by 10, 15, 21 and 28. After that the common difference between the number and its respective remainder is subtracted from that i.e. 9660 – 6 = 9654. Hence 9654 will be the answer for this question.

Ex. 8: What is the greatest number that will divide 2400 and 1810 and leaves remainders 6 and 4 respectively?

Sol: Since on dividing 2400, remainder 6 is left, the required number must divide (2400 – 6) or 2394 exactly. Similarly, it must divide must divide (1810 – 4) or 1806 exactly. Hence, the greatest required number should be the HCF of 2394 and 1806, i.e. 42.

Ex. 9: What is the greatest number that will divide 38, 45 and 52 and leaves remainders as 2, 3 and 4 respectively?

Sol: The required greatest number will be the HCF of (38 – 2), (45 – 3) and (52 – 4) or 36,42 and 48.
∴Answer is 6.

Ex. 10: Find the greatest number which will divide 410,751 and 1030 so as to leave remainder 7 in each case.

Sol: The required greatest number = HCF of (410 – 7), (751 – 7) and (1030 – 7).
∴ Answer is 31.

Ex. 11: Find the greatest number which is such that when 76,151 and 226 are divided by it, the remainders are all alike. Find also the common remainder.

Sol: Let k be the remainder, then the numbers (76 – k), (151 – k) and (226 – k) are exactly divisible by the required number. Now, we know that if two numbers are divisible by a certain number, then their difference is also divisible by that number.

Hence, the numbers (151 – k) - (76 – k),(226 – k) - (151 – k) and (226 – k) - (76 – k) or 75,75 and 150 are divisible by the required number.
Therefore, the required number = HCF of75, 75 and150 = 75.
And the remainder will be found after dividing 76 by 75 as 1.

Ex. 12: The numbers 11284 and 7655, when divided by a certain number of three digits, leave the same remainder. Find that number of three digits.

Sol: The required number must be a factor of (11284 – 7655) or3629.Now, 3629 = 19 ×191
∴ 191 is the required number.

Ex. 13: The product of two numbers is 7168 and their HCF is16; find the numbers.

Sol: The numbers must be multiples of their HCF. So, let the numbers be 16a and16b where a and b are two numbers prime to each other. ∴16a × 16b = 7168 or, ab = 28
Now, the pairs of numbers whose product is 28 are (28,1);(14,2);(7,4).
14 and 2 which are not prime to each other should be rejected.
Hence, the required numbers are 28 ×16, 1×16;7×16,4×16 or ,448,16;112,64.

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Ex. 14: The LCM of two numbers is 2079 and their HCF is 27. If one of the numbers is 189, find the other.

Sol: The required number = $\frac{LCM \times HCF}{FirstNumber} = \frac{2079 \times 27}{189} = 297.$

Ex. 15: Find the least number which, when divided by 18, 24, 30 and 42, will leave in each case the same remainder 1.

Sol: Clearly, the required number must be greater than the LCM of 18, 24, 30 and 42 by 1.
Now, $18 = 2 \times 3^2$, $24 = 2^3 \times 3$, $30 = 2 \times 3 \times 5$, $42 = 2 \times 3 \times 7$.
$\therefore$ LCM = $3^2 \times 2^3 \times 5 \times 7 = 2520$
$\therefore$ The required number = 2520 + 1 = 2521.

Ex. 16: What is the least number which, when divided by 52, leaves 33 as the remainder, and when divided by 78 leaves 59, and when divided by 117 leaves 98 as the respective remainders?

Sol: Since 52 – 33 = 19,78 – 59 = 19, 117 – 98 = 19, we see that the remainder in each case is less than the divisor by 19.
Hence, if 19 is added to the required number, it becomes exactly divisible by 52,78 and 117. Therefore, the required number is19 less than the LCM of 52,78 and 117.
The LCM of 52,78 and 117 = 468
$\therefore$ The required number = 168 -19 = 449.

Ex. 17: Find the greatest number of six digits which, on being divided by 6,7,8,9 and 10, leaves 4,5,6,7 and 8 as remainders respectively.

Sol: The LCM of 6,7,8,9 and 10 = 2520. The greatest numbers of 6 digits = 999999.
Dividing 999999 by 2520, we get 2079 as remainder.
Hence, the 6 - digit number divisible by 2520 is (999999-2079), or 997920.
Since 6 -4 = 2,7-5 = 2,8- 6 = 2,9-7 = 2,10 – 8 = 2,the remainder in each case is less than the divisor by 2.

Ex. 18: Find the greatest number less than 900, which is divisible by 8,12 and 28.

Sol: The least number divisible by 8, 12 and 28 is 84. Clearly, any multiple of 84 will be exactly divisible by each of the numbers 8, 12 and 28. But since the required number is not to exceed 900, it is 84×10 = 840.

Ex. 19: Find the least number which, upon being divided by 2, 3, 4, 5, 6 leaves in each case a remainder of 1, but when divided by 7 leaves no remainder.

Sol: The LCM of 2, 3, 4, 5, 6 = 60.
$\therefore$ The required number must be = 60k + 1, where k is a positive integer. = (7 × 8 + 4)k + 1= (7 × 8k) + (4k + 1).
Now, this number is to be divisible by 7. Whatever may be the value of k, the portion (7 × 8k) is always divisible by 7.
Hence, we must choose that least value of k which will make 4k + 1 divisible by 7.
Putting k = 1, 2, 3, 4, 5 etc. in succession, we find that k should be 5.
$\therefore$ The required number = 60k + 1 = 60 × 5 + 1 = 301.

Ex. 20: What least number must be subtracted from 1936 so that the remainder when divided by 9,10,15 will leave in each case the same remainder 7?

Sol: The LCM of9, 10 and 15 = 90. On dividing 1936 by 90, the remainder = 46.
But 7 is also a part of this remainder
$\therefore$ The required number = 46 – 7= 39.

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Ex. 21: What greatest number can be subtracted from10,000 so that the remainder may be divisible by 32, 36, 48 and 54?

Sol: LCM of 32, 36,48,54 = 864. ∴ The required greatest number = 10,000 – 864 = 9136.

Ex. 22: What is the least multiple of 7, which when divided by 2, 3, 4, 5 and 6, leaves the remainders 1, 2, 3, 4, 5 respectively?

Sol: LCM of 2, 3, 4, 5, 6 = 60. Other numbers divisible by 2, 3, 4, 5, 6 are 60k, where k is a positive integer. Since 2 – 1 = 1, 3 – 2 = 1, 4 – 3 = 1, 5 – 4 = 1 and 6 – 5 = 1, the reminder in each case is less than the divisor by 1, the required number = 60k – 1 = (7 × 8k) + (4k – 1)
Now, this number is to be divisible by 7. Whatever may be the value of k the portion 7 × 8k is always divisible by 7. Hence, we must choose the least value of k which will make (4k – 1) divisible by 7. Putting k equal to 1, 2, 3, etc. in succession, we find that k should be 2.
∴The required number = 60k – 1 = 60 × 2 – 1 = 119.

Ex. 23: What is the HCF and LCM of $\frac{4}{5}, \frac{5}{6}$, and $\frac{7}{15}$?

Sol:

$$HCF = \frac{HCF\ \ of\ \ Numerators}{LCM\ \ of\ \ Deno\min ators} = \frac{1}{30}$$

$$LCM = \frac{LCM\ \ of\ \ Numerators}{HCF\ \ of\ \ Deno\min ators} = \frac{140}{1} = 140$$

Ex. 24: Three men start together to travel the same way around a circular track of 11 km. Their speeds are 4, 5.5 and 8 km per hour respectively. When will they meet at the starting point?

Sol: Time taken by them to complete one revolution

$$= \frac{11}{4}, \frac{11}{5.5}\ \ and\ \ \frac{11}{8} hrs\ \ respectivedy = \frac{11}{4}, \frac{2}{1}\ \ and\ \ \frac{11}{8}$$

$$LCM\ \ of\ \ \frac{11}{4}, \frac{2}{1}\ \ and\ \ \frac{11}{8} = \frac{LCM\ of\ 11, 2, 11}{HCF\ \ of\ \ 4, 1, 8} = \frac{22}{1} = 22hrs.$$

∴They will meet after 22 hrs.

Ex. 25: Find the smallest whole number which is exactly divisible by -

$$1\frac{1}{2}, 1\frac{1}{3}, 2\frac{1}{4}, 3\frac{1}{2}\ \ and\ \ 4\frac{1}{5}$$

Sol: The required smallest number = LCM of the given numbers. The given numbers are -

$$\frac{3}{2}, \frac{4}{3}, \frac{9}{4}, \frac{7}{2}, \frac{21}{5}$$

Using the formula for LCM of fractions, we get the answer as -

$$\frac{252}{1} = 252.$$

- Whenever the question talks about the greatest or maximum, then in most of the cases it will be a question of HCF. Secondly whenever the question is related to classification or distribution into groups then in all the cases it is HCF only.
- Whenever the question talks about the smallest or minimum, then in most of the cases it will be a question of LCM. Secondly whenever the word 'together' or 'simultaneous' in used in the question then in all the cases it is LCM.

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Averages

Average means Arithmetic mean of the items and it is = $\frac{\text{Sum of Items}}{\text{Number of Items}}$

When the difference between all the items is same, then average is equal to $\frac{n+1}{2}$ item, where *n* is the total number of items.

Average speed: If a man covers some journey from A to B at *u* km/hr. and returns back to A at B uniform speed of *v* km/hr., then the average speed during the whole journey is $\frac{2uv}{u+v}$ km/hr

- If *x* is added in all the items, then average increases by *x*.
- If *x* is subtracted from all the items, then average decreases by *x*.
- If every item is multiplied by *x*, then average also gets multiplied by *x*.
- If every item is divided by *x*, then average also gets divided by *x*.
- This means if average increases by y, it can be assumed that y is added in all the items.

5.1 Solved Examples based on Average:

Ex. 1: Find the average of first three prime numbers.

Sol: First three prime numbers are 2, 3 and 5. Their average is $\frac{2+3+5}{3} = \frac{10}{3}$ = 3.33.

Ex. 2: Find the average of 8, 11, 14, 17 and 20.

Sol: The sum of the items is 8 + 11 + 14 + 17 + 20. Average = $\frac{70}{5}$ = 14.

Ex. 3: Find the average of 18, 21, 24, 27 & 30.

Sol: Here in this question, the difference between all the items is same, therefore average would be equal to (n + 1) ÷ 2 item. (5 + 1)/2 = 3rd item i.e. 24 would be average of these five items. As in the Ex. 2 above, the difference between all the items was same and there are 5 items then average would be equal to (5+1)/2 item i.e. the 3rd item, which was given to be 14 was the answer.

Ex. 4: The average of seven consecutive numbers is 47. Find the smallest of these numbers.

Sol: As the numbers given are consecutive, their difference would always be equal. Average would be equal to (7+1) ÷ 2 item means the 4th item, which is given to be 47. This means 47 is the 4th number, 46 is the 3rd number, 45 is the 2nd number, 44 is the 1st and the smallest number.

Ex. 5: The average temperature for Monday, Tuesday and Wednesday was 39.8°C. The average for Tuesday, Wednesday and Thursday was 40.8°C. If the temperature on Monday be 38°C, find the temperature on Thursday.

Sol: In this question, the sum of three days Monday, Tue and Wed is 39.8 × 3 = 119.4°
Similarly the sum of Tuesday, Wednesday, Thursday is 40.8 × 3 = 122.4°
Here whatever is the difference between the two totals it must be due to Monday and Thursday because the remaining two days are common in both of these cases.
The difference is 3°, which means Thursday must be 3 more than Monday because total in case of Thursday is more. Now the temperature on Monday is given to be 38°, therefore the temperature on Thursday is 38 + 3 = 41°.
But there is no need to do all this, a straight logic can be applied, which says if average has increased by 1°, it can be assumed that 1 is added in all the items and there are 3 items. This means 3 is added in the total, which must be difference between Monday and Thursday (the uncommon days). Even in this manner the same difference of 3 can be calculated.

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Then similarly 3 would be added in the temperature of Monday to calculate temperature on Thursday.
You can follow the following process.
1. Firstly take up the difference between the two averages.
2. Multiply it with the number of items.

Ex. 6: The average of 30 boys of a class is equal to 14 yrs. When the age of the class teacher is included the average becomes 15 yrs. Find the age of the class teacher.

Sol: Total ages of 30 boys = 14 × 30 = 420 yrs. Total of age when class teacher is included = 15 × 31 = 465 yrs. ∴ Age of class teacher = 465 – 420 = 45 yrs.

Ex. 7: The average weight of 4 men is increased by 3 kg when one of them who weighs 120 kg is replaced by another man. What is the weight of the new man?

Sol: If the average is increased by 3 kg, then the sum of weights increases by 3 × 4 = 12 kg.
And this increase in weight is due to the extra weight included due to the inclusion of new person.
∴ Weight of new man = 120 + 12 = 132 kg.

Ex. 8: The average of marks obtained by 120 candidates in a certain examination is 35. If the average marks of passed candidates are 39 and that of the failed candidates is 15, what is the number of candidates who passed the examination?

Sol: Let the number of passed candidates be x. Then total marks = 120 × 35 = 39x + (120 – x) × 15
Or, 4200 = 39x + 1800 – 15x or, 24x = 2400 ∴ x = 100 ∴ number of passed candidates = 100.

Ex. 9: The average of 11 results is 50. If the average of first six results is 49 and that of last six is 52, find the sixth result.

Sol: The total of 11 results = 11 × 50 = 550. The total of first 6 results = 6 × 49 = 294
The total of last 6 results = 6 × 52 = 312. The 6th result is common to both;
∴ Sixth result = 294 + 312 – 550 = 56

Ex. 10: The average age of 8 persons in a committee is increased by 2 years when two men aged 35 yrs and 45 yrs are substituted by two women. Find the average age of these two women.

Sol: The total age of two women = 2 × 8 + (35 + 45) = 16 + 80 = 96 yrs.
∴ Average age of two women = $\frac{96}{2} = 48\text{yrs}$.

Ex. 11: The average age of a family of 6 members is 22 yrs. If the age of the youngest member be 7 yrs, then what was the average age of the family at the birth of the youngest member?

Sol: Total ages of all members = 6 × 22 = 132 yrs. 7 yrs ago, total sum of ages 132 – (6 × 7) = 90 yrs. But at that time there were 5 members in the family.
∴ Average at that time = 90 ÷ 5 = 18 yrs.

Ex. 12: A man bought 13 shirts of Rs 50 each, 15 pants of Rs 60 each and 12 pairs of shoes at Rs 65 a pair. Find the average value of each article.

Sol: Average = $\frac{13\times 50+15\times 60+12\times 65}{13+15+12} = \text{Rs.}58.25$

Ex. 13: The average score of a cricketer in two matches is 27 and in three other matches is 32. Then find the average score in all the five matches.

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Sol: Average in 5 matches = $\frac{2\times 27+3\times 32}{2+3}=\frac{54+96}{5}=30$

Ex. 14: The average of 11 results is 30, that of the first five is 25 and that of the last five is 28. Find the value of the 6th number.

Sol. 6th number = Total of 11 results – (Total of first five + Total of last five results)
= 11 × 30 – (5 × 25 + 5 × 28) = 330 – 265 = 65

Ex. 15: A batsman in his 17th innings makes a score of 85, and thereby increases his average by 3. What is his average after 17 innings?

Sol. Let the average after 16th innings be x, then 16x + 85 = 17 (x + 3) = Total score after 17th innings.
∴ x = 85 – 51 = 34∴average after 17 innings = x + 3 = 34 + 3 = 37

Average related to speed

- If a person travels a distance at a speed of x km/hr and the same distance at a speed of y km/hr, then the average speed during the whole journey is given by $\frac{2xy}{x+y}\text{km/hr}$.

  Or,
- If half of the journey is travelled at a speed of x km/hr and the next half at a speed of y km/hr, then average speed during the whole journey is $\frac{2xy}{x+y}\text{km/hr}$.

  Or,
- If a man goes to a certain place at a speed of x km/hr and returns to the original place at a speed of y km/hr, then the average speed during up-and-down journey is $\frac{2xy}{x+y}\text{km/hr}$.

- If a person travels three equal distances at a speed of x km/hr, y km/hr and z km/hr respectively, then the average speed during the whole journey is $\frac{3xyz}{xy+yz+xz}\text{km/hr}$.

Ex. 16: A train travels from A to B at the rate of 20 km per hour and from B to A at the rate of 30 km/hr. What is the average rate for the whole journey?

Sol: Required Average speed = $\frac{2\times 20\times 30}{20+30}=24\text{km/hr}$.

Ex. 17: A person divides his total route of journey into three equal parts and decided to travel the three parts with speeds of 40, 30 and 15 km/hr respectively. Find his average speed during the whole journey.

Sol. Required Average speed = $\frac{3\times 40\times 30\times 15}{40\times 30+30\times 15+40\times 15}=\frac{3\times 40\times 30\times 15}{2250}=24km/hr.$

Ex. 18: One-third of a certain journey is covered at the rate of 25 km/hr, one-fourth at the rate of 30 km/hr and the rest of 50 km/hr. Find the average speed for the whole journey.

Sol: Let the total journey be x km. Then $\frac{x}{3}$ km at the speed of 25 km/hr and $\frac{x}{4}$ km at 30 km/hr and the remaining distance $\left(x-\frac{x}{3}-\frac{x}{4}\right)\frac{5}{12}x$ at the speed of 50 km/hr.

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Total time taken during the journey of x km

$$= \frac{x}{3\times 25} hrs + \frac{x}{4\times 30} hrs + \frac{5x}{12\times 50} hrs = \frac{18x}{600} hrs = \frac{3x}{100} hrs$$

$\therefore$ Average speed = $\dfrac{x}{\frac{18x}{600}} = \dfrac{100}{3} = 33\dfrac{1}{3} km/hr.$

Ex. 19: The average salary of the entire staff in a office is Rs 120 per month. The average salary of officers is Rs 460 and that of non-officers is Rs 110. If the number of officers is 15, then find the number of non-officers in the office.

Sol. Let the required number of non-officers = x.
Then, 110x + 460 × 15 = 120 (15 + x)
∴ x = 15 × 34 = 510.

Ex. 20: There were 35 students in a hostel. If the number of students increases by 7, the expenses of the mess increase by Rs 42 per day while the average expenditure per head diminishes by Re 1. Find the original expenditure of the mess.

Sol: Suppose the average expenditure was Rs x. Then total expenditure = 35x.
When 7 more students join the mess, total expenditure = 35x + 42.

Now, the average expenditure = $\dfrac{35x+42}{35+7} = \dfrac{35x+42}{42}$

Now, we have $\dfrac{35x+42}{42} = x - 1$ or, 35x + 42 = 42x – 42or, 35x + 42 = 42x – 42
or, 7x = 84 ∴ x = 12.
Thus the original expenditure of the mess = 35 × 12 = Rs. 420.

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